

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12398975
Kind Code	B1
Date of Patent	August 26, 2025
Inventor(s)	Butterfield; Ethan T.

Trigger assembly for air gun

Abstract

A trigger assembly includes an actuator that moves along a longitudinal axis, a primary sear, a catch, a trigger, and a secondary sear. After releasing the actuator to fire the air gun, the actuator engages the primary sear, which cooperates with the secondary sear and the catch to facilitate automatic re-cocking of the air gun.

Inventors:	Butterfield; Ethan T. (Ontario, NY)
Applicant:	Crosman Corporation (Bloomfield, NY)
Family ID:	1000007491423
Assignee:	Crosman Corporation (Bloomfield, NY)
Appl. No.:	18/504956
Filed:	November 08, 2023

Publication Classification

Int. Cl.: F41B11/00 (20130101); F41B11/62 (20130101); F41B11/70 (20130101)

U.S. Cl.:

CPC F41B11/70 (20130101); F41B11/62 (20130101);

Field of Classification Search

CPC: F41B (11/646); F41B (11/70); F41B (11/62); F41A (17/46); F41A (19/16); F41A (19/33); F41A (19/56); F41A (3/40); F41A (3/76); F41A (7/06)

USPC: 24/67; 124/37; 124/71; 124/40; 124/48; 124/51.1; 89/1.1; 89/152; 42/75.03

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
861939	12/1906	Benet et al.	89/142	F41A 19/33
1294892	12/1918	Fox et al.	89/33.02	F41A 3/40
1396404	12/1920	Burton	124/40	F41B 11/647
1886561	12/1931	Lefever	124/67	F41B 11/646
3464399	12/1968	Hordley	124/40	F41B 11/646
3561319	12/1970	Joyce	42/59	F41A 9/23
4899717	12/1989	Rutten	124/66	F41B 11/71

Primary Examiner: David; Michael D

Attorney, Agent or Firm: Lee & Hayes, P.C.

Background/Summary

BACKGROUND

(1) Air guns are used for a variety of recreational purposes. Some conventional air guns include a tank or reservoir that is charged (e.g., pre-charged) to provide a number of shots before re-charging. These pre-charged pneumatics (PCPs) are able to supply air “on-demand,” e.g., for ready firing. Because compressed air is readily available on PCPs, those air guns may be configured for rapid firing, such as semi-automatic firing. Thus, there is a need in the art for an improved trigger assembly, which may be used with a PCP or other type of on-demand air gun and that facilitates self-cocking and/or semi-automatic firing of projectiles.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The detailed description is described with reference to the accompanying figures. In the figures, the left-most digit of a reference number identifies the figure in which the reference number first appears. The same reference numbers in different figures indicate similar or identical items.

(2) FIG. 1 is side view of an example air gun according to example implementations of this disclosure.

(3) FIGS. 2A through 2H are side elevation views of aspects of a trigger assembly illustrating steps of cocking and (automatic) re-cocking of the trigger assembly according to example embodiments of the present disclosure.

DETAILED DESCRIPTION

(4) This application relates to a trigger assembly for use with an air gun or other projectile launching device. In examples, trigger assemblies disclosed herein may be used in a pre-charged pneumatic (PCP) air gun. The systems and techniques described herein may provide for automatic re-cocking of a trigger assembly, e.g., to facilitate semi-automatic firing of the air gun including the trigger assembly.

(5) In aspects of this disclosure, a trigger assembly includes a trigger, a primary sear configured to rotate about a first pivot, a secondary sear configured to rotate about a second pivot, and a catch configured to rotate about a third pivot. During cocking of the air gun, a firing actuator, such as a

hammer, is moved to a cocked position, e.g., against a biasing force of a firing spring. In the cocked position, the primary sear engages the hammer, and the secondary sear is positioned to resist a force applied on the primary sear by the firing spring, e.g., via the hammer. From the cocked position, the trigger assembly is ready for firing.

(6) The trigger is pulled to fire the air gun. In examples, the trigger includes a trigger extension or similar surface that contacts the secondary sear, causing the secondary sear to rotate, out of engagement with the primary sear. Once the secondary sear is disengaged from the primary sear, the primary sear is no longer able to retain the hammer in the cocked position. The hammer advances in a firing direction under the biasing force of the firing spring, moving the primary sear out of a path of travel of the hammer. Continued movement of the hammer causes the hammer to impact a valve stem or similar firing member that results in a rush of compressed air into a firing chamber, to fire a projectile from the air gun.

(7) After firing, the hammer recoils. In examples, the force of the recoil is sufficient to compress the firing spring such that the hammer returns to a position corresponding to the seared or cocked position, e.g., a re-seared or re-cocked position. In this re-seared or re-cocked position, the primary sear again blocks advancement of the hammer, and the catch, separate from the secondary sear, engages the primary sear to counter the biasing force applied to the primary sear by the hammer/firing spring. For example, the trigger is retained in the actuated, pulled, or firing position such that the secondary sear remains disengaged from the primary sear during the recoil and re-searing or re-cocking.

(8) When the trigger is released, a trigger return spring biases the trigger back to the unpulled or normal state. This return of the trigger causes the secondary sear to return to a position to support the primary sear. The return of the trigger also causes the catch to disengage from the primary sear. The result is that, with the trigger fully returned to the neutral or unfired position, the trigger assembly is arranged in the cocked position, and ready to fire another projectile.

(9) Thus, systems and techniques described herein may facilitate automatic re-cocking of the trigger assembly. For example, aspects of this disclosure describe arrangements in which a firing actuator is re-seared automatically after firing, when the trigger is retained in the pulled or firing position. For example, the catch may retain the primary sear in a blocking position, e.g., to maintain the hammer in the cocked position, until the trigger is returned to the firing position. The return of the trigger to the firing position causes a “handoff” of the primary sear from the catch to the secondary sear, to place the trigger assembly in the cocked position.

(10) Specific examples of this disclosure are provided below with reference to the figures.

Although the specific examples described reference use of a trigger assembly with a PCP air gun, aspects of this disclosure can be used with other types of air guns. Without limitation, the trigger assemblies described herein may provide action for any number of projectile launchers.

(11) FIG. 1 illustrates an example PCP air gun **100** according to aspects of this disclosure. More specifically, FIG. 1 is a side view of one implementation of the air gun **100**, which includes a trigger assembly for self-cocking and firing via an improved trigger assembly. FIG. 1 illustrates the air gun **100** as generally including a barrel **102**, a stock **104**, and a trigger **106**. The air gun **100** also includes a housing **108** extending generally between the barrel **102** and the stock **104**. The housing **108** may retain and/or conceal components of the air gun **100**, including a trigger assembly, which includes the trigger **106**. Without limitation, aspects of this disclosure include components for self- or automatic-cocking of the air gun **100** using the trigger assembly, which may be disposed in, attached to, or otherwise associated with the housing **108**. In aspects of this disclosure, the housing **108** may be any portion of the air gun **100** that contains, retains, mounts, or otherwise couples to other aspects of the air gun. In some examples, the housing may be a portion of the air gun **100** that conceals, covers, and/or shrouds other aspects of the air gun. For example, the housing **108** can include some or all of the stock **104**, the barrel **102**, and/or a central body of the air gun between the stock **104** and/or the barrel.

(12) The barrel **102** extends generally from a breech end **110** to a muzzle end **112**. Although not illustrated in FIG. **1**, a bore extends through the barrel **102**, from the breech end **110** to the muzzle end **112**. The bore provides a hollow interior space within the barrel **102** through which compressed air and a projectile, such as a pellet, can pass, as will be described in greater detail below. The barrel **102** is sufficiently strong to contain high pressure gasses introduced into the barrel **102** to fire the projectile. In implementations, the bore may be smooth, or the bore may be rifled, e.g., to impart a stabilizing spin on the projectile as it passes through the bore.

(13) The stock **104** may be any conventional size or shape. In some instances, the stock **104** may be removably secured to the housing, e.g. to promote removal and/or replacement of the stock **104**. Moreover, removal of the stock **104** may facilitate access to an interior of the housing **108**, e.g., to service working components of the air gun **100**. The stock **104** may also house a battery or other power source, a control system adapted to receive inputs from sensors and/or to generate outputs that drive motors, actuators, lights, solenoids, motors, linear actuators and/or other electronic, mechanical, electromechanical, sonic, electrooptical, and/or other components. Without limitation, the stock **104** may comprise a portion of the housing **108**.

(14) The trigger **106** may be any lever, button, or the like, configured for user interaction to fire the air gun **100**. As detailed further herein, in some instances the trigger **106** is a part of a trigger assembly that, among other features, facilitates semi-automatic firing and/or self-cocking of the air gun **100**. A trigger assembly including the trigger **106** may also or alternatively prevent firing of the air gun **100** while the air gun **100** is presenting a new projectile for firing, e.g., after firing a projectile.

(15) The housing **108** is generally provided to contain components of the air gun **100**. For instance, and as detailed further below, the housing **108** may contain, support, and/or conceal aspects that facilitate action of the air gun **100**. The shape and size of the housing **108** in FIG. **1** is for illustration. Other shapes, sizes, and compositions are contemplated. Components of the housing may be made of any conventional materials, including but not limited to, metal, such as aluminum, or polymers.

(16) FIG. **1** also illustrates that the stock **104** can include one or more removable sections **114**, **116**. For example, the removable sections **114**, **116** can be movable relative to the stock **104** to provide access to an interior of the air gun **100**. Without limitation, the can include a user interface **138**. The user interface **138** is illustrated as including a number of buttons, switches, or similar interface elements that may be used to control aspects of the air gun **100**. An example of the user interface **138** is illustrated in FIG. **4**, described in more detail below.

(17) FIGS. **2-9** are side views of aspects of a trigger assembly **200** that may be incorporated in the air gun **100**. For example, the trigger assembly **200** may facilitate automatic cocking of the air gun **100**, e.g., to allow for semi-automatic firing of the air gun **100**. More specifically, FIGS. **2-9** show a progression of cocking, firing and re-cocking of the air gun **100**, e.g., automatically, to enable consecutive firing of the air gun **100** without (or with minimal) user interaction.

(18) The trigger assembly **200** is generally configured to work with a conventional pre-charged pneumatic air gun, as described further herein. Although not illustrated in FIG. **2**, the air gun **100** includes a source of compressed air, e.g., a pre-charged canister (such as a compressed air canister), a cartridge (such as a CO2 cartridge), or the like. In operation, generally, the compressed air is provided to the barrel **102**, e.g., via a conduit **202** (shown for example only). Not illustrated in FIG. **2**, the air gun **100** further includes valves and/or other controllable components configured to allow a portion of compressed air to escape down the barrel **102**, forcing a pellet or other projectile (not illustrated) arranged in the barrel **102** to be fired from the barrel **102**. The trigger assembly **200** is configured to facilitate the aforementioned escape of compressed air down the barrel **102**, e.g., as a shot. As detailed further herein, the trigger assembly **200** also is configured to ready the air gun **100** for firing of additional shots, e.g., automatically and without additional user manipulation, to facilitate semi-automatic firing.

(19) The trigger assembly **200** includes the trigger **106** and a number of additional components that cooperate to fire and re-cock the air gun **100**. Specifically, FIG. **2** shows an actuator **204**, a primary sear **206**, a secondary sear **208**, and a catch **210** forming portions of the trigger assembly.

(20) The actuator **204** is configured to move generally along, e.g., parallel to a firing axis **212**. Without limitation, the firing axis **212** may be aligned with an axis of the barrel **102** and thus may be an axis along which projectiles are fired. In examples, the actuator **204** may be a hammer, although in other arrangements the actuator **204** may be at least partially formed by some other component. Without limitation, aspects of the actuator **204** may be incorporated into a valve assembly, e.g., used to fire the air gun **100**.

(21) As shown, the actuator **204** extends between a leading end **214**, e.g., toward the barrel **102**, and a trailing end **216**. The actuator **204** may have a cylindrical sidewall, although other shapes and/or configurations are also contemplated. Moreover, and as detailed further herein, the actuator **204** can include one or more additional features, including but not limited to an engagement surface **218** offset, e.g., longitudinally, from the leading end **214** and the trailing end **216**. The engagement surface **218** may cause the outer surface, e.g., a sidewall, of the actuator **204** to have a contour or surface features. The leading end **214**, the trailing end **216**, and the engagement surface **218** are detailed further below.

(22) The primary sear **206** is a generally elongate member, extending between a first, leading end **220** and a second, trailing end **222**. The primary sear **206** includes a first protrusion **224** proximate the leading end **220**. The first protrusion **224** extends generally upward (in the orientation of FIG. **2A**) toward the actuator **204**. The primary sear **206** also includes a second protrusion **226** proximate the trailing end **222**. The second protrusion **226** extends generally upward (in the orientation of FIG. **2A**) toward the actuator **204**. As also shown in FIG. **2A**, the primary sear **206** also includes a catch engagement **228** proximate the trailing end **222**. The catch engagement **228** is illustrated as including a contoured surface proximate a lower (in the orientation of FIG. **2A**) side of the body of the primary sear **206**. As also illustrated in FIG. **2A**, the primary sear **206** is configured to rotate about a pivot **230**. More specifically, and as detailed herein, aspects of the primary sear **206**, including the leading end **220**, the first protrusion **224**, the second protrusion **226**, and/or the catch engagement **228** are configured to selectively cooperate with one or more of the actuator **204**, the secondary sear **208** and/or the catch **210**. For example, pivoting the primary sear **206** about the pivot **230** selectively positions aspects of the primary sear **206** to interface, interact, or otherwise cooperate with components of the trigger assembly **200**, as detailed further herein.

(23) The secondary sear **208** includes a generally elongate body and a secondary sear surface **232** extending above the elongate body. The secondary sear **208** is configured to rotate about a second pivot **234** spaced from the first pivot **230**. As detailed further herein, the secondary sear surface **232** may be configured to contact the leading end **220** of the primary sear **206** in some orientations of the trigger assembly **200**, as discussed further herein. More specifically, in some examples the primary sear **206** may be configured to retain the actuator **204** in a cocked position and the secondary sear **208** may be configured to retain the primary sear **206** in a desired position.

(24) The catch **210** is configured to rotate about a third pivot **236**, which is spaced from the first pivot **230** and the second pivot **234**. As illustrated, the catch **210** includes a hook **238** spaced from the third pivot **236**. As detailed further herein, the hook **238** is configured to cooperate with the catch engagement **228** on the primary sear **206**. For example, the hook **238** may engage with the catch engagement **228** with the trigger assembly in a cocked position, e.g., to retain the primary sear **206** in a position to hold the actuator against a biasing force to fire the air gun **100**.

(25) The catch **210** also includes a cammed, trigger engagement surface **240**, at an end of the catch **210** opposite the hook **238**. As detailed further herein, the trigger engagement surface **240** can engage a trigger surface **242** on the trigger **106**, e.g., during actuation of the trigger **106** by a user, or, and as detailed further below, during release of the trigger **106**.

(26) As also illustrated, the trigger **106** can include a trigger extension **243**. In some

implementations, the trigger extension **243** may be a protrusion that extends upward (in the orientation of FIG. 2A). For example, the trigger extension **243** may be a columnar member having a distal end extending a height above a portion of the trigger **106**, e.g., above the trigger surface **242**. As detailed further herein, the trigger extension **243** may be positioned to contact the secondary sear **208** when the trigger **106** is pulled, e.g., to cause the secondary sear **208** to disengage the primary sear **206**. In examples, a height of the trigger extension may be adjustable to alter aspects of the trigger assembly **200**. For example, reducing the height of the trigger extension **243** may require a longer trigger pull to fire the air gun **100**.

(27) FIG. 2A also illustrates that the air gun **100** further includes a firing spring **244**, proximate the trailing end **216** of the actuator **204**. As detailed further herein, the firing spring **244** is configured to bias the actuator **204** toward the barrel **102**, e.g., to fire the air gun **100**. FIG. 2A also shows a valve stem **246**, proximate the leading end **214** of the actuator **204**. Application of a force on the valve stem **246** generally in the firing direction illustrated in FIG. 2A causes compressed air to enter the barrel **102**, e.g., via the conduit **202**, which air escapes down the barrel **102** to fire a projectile positioned in the air flow. Although not illustrated in FIG. 2A, the valve stem **246** is biased to a rearward position (e.g., the rearward position illustrated), but the compressed air and/or by a biasing member associated with the barrel **102** or an action mechanism associated with the barrel **102**. Thus, and as detailed further herein, to fire the air gun **100**, the actuator **204** may be forced in the firing direction by a force from the firing spring **244**. Under this force, the actuator **204** contacts the valve stem **246**, causing a valve associated with the barrel **102** to open, e.g., momentarily, to allow firing the air gun **100**.

(28) In addition to the components and features just discussed and referenced in FIG. 2A, other features and functionality of the trigger assembly, as well as some modifications to these features and functionality, will be introduced and detailed further herein.

(29) In the example of FIG. 2A, the trigger assembly **200** is in a normal or discharged configuration. For example, in the normal configuration, the trigger assembly **200** is not cocked and is not firing. The normal configuration may be a configuration in which the air gun **100** is stored, a configuration in which the air gun is maintained, e.g., to recharge the air gun and/or to load projectiles to the air gun, and/or the like. In the normal configuration, movement of the actuator **204** along the axis **212**, e.g., between the firing spring **244** and the valve stem **246** may be substantially unimpeded. Also in the normal configuration, the trigger **106** is unactuated or in a neutral position. Pulling the trigger in the normal configuration may cause no appreciable action at the air gun **100**.

(30) FIGS. 2B-2H show a sequence of configurations of the trigger assembly **200** during use of the air gun **100**. Throughout these figures, the same reference numbers refer to the same components, unless otherwise noted.

(31) FIG. 2B shows a cocking configuration, in which the air gun is being cocked from the normal configuration shown in FIG. 2A. Specifically, in FIG. 2B, the actuator **204** is moved against the biasing force of the firing spring **244** to compress the firing spring **244**. Specifically, the actuator **204** is moved generally in the direction of an arrow **248** (opposite the firing direction shown in FIG. 2A) such that the trailing end **216** of the actuator **204** abuts the firing spring **244**. Continued movement of the actuator **204** in the direction causes the actuator **204** to compress the firing spring **244**. For example, and although not illustrated in the FIGS., the air gun **100** may further include a manual interface, e.g., as a bolt, a slide, or the like, associated with the actuator **204**. A user may interact with the manual interface to move the actuator **204** into the position shown in FIG. 2B. In at least some examples, the movement of the actuator **204** may also facilitate loading of a projectile into a firing position in or proximate the barrel **102**. In examples, the trigger assembly **200** and associated functionality may be used with any manner of loading projectiles, including manual, semi-automatic, or automatic feeding. In some examples, the cocking force required to overcome the biasing force of the firing spring **244** may be provided by an automatic cocking mechanism,

such as an electromagnetic slide, or the like.

(32) In addition to compressing the firing spring **244**, and as shown in the highlighted section **250** of FIG. **2B**, movement of the actuator **204** along the direction of the arrow **248** also causes the engagement surface **218** to engage the second protrusion **226** of the primary sear **206**. As illustrated, the engagement surface **218** is an angled surface, and the second protrusion **226** includes a complementary angled surface. As illustrated in FIG. **2B**, the contact of the engagement surface **218** with the second protrusion **226** causes the primary sear **206** to pivot about the first pivot **230**, e.g., in counterclockwise direction. As a result of this pivoting, the first protrusion **224** of the primary sear **206** moves upward. As shown, the primary sear **206** may pivot such that the first protrusion **224** is disposed in an upward position in a path of travel of the actuator **204**, e.g., should the actuator **204** move in the firing direction.

(33) FIG. **2C** shows the trigger assembly **200** in a cocked position. Specifically, in the example of FIG. **2C**, the (cocking) force applied in the example of FIG. **2B** to overcome the bias of the firing spring **244** is removed. As noted, during cocking, the primary sear **206** pivots such that the first protrusion **224** (at the leading edge **220** of the primary sear **206**) is in a path of travel of the actuator **204**. With the cocking force removed, the force of the firing spring **244** biases the actuator **204** in the firing direction. As shown in the highlighted section **252** of FIG. **2C**, the leading surface **214** of the actuator **204**, under the force of the firing spring **244**, contacts the first protrusion **224** to maintain the actuator **204** against the spring force of the firing spring **244**.

(34) Moreover, and as also illustrated in the highlighted portion **250**, the secondary sear **208** is positioned such that the secondary sear surface **232** is disposed under the leading end **220** of the primary sear **206**. As detailed above, the secondary sear **208** is configured to rotate about the second pivot **234**. In this example, the force transferred from the primary sear **206** to the second sear, e.g., at the secondary sear surface **232** is substantially aligned with the pivot **234**, such that minimal rotational force is applied to the secondary sear **208**, e.g., about the second pivot **234**. In examples, a biasing spring or the like may be sufficient to offset any rotational force resulting from contact of the primary sear **206** with the secondary sear **208**.

(35) The catch **210** may also assist in maintaining the primary sear **206** in the illustrated cocked position. Specifically, and as shown in the highlighted section **254**, an outer surface **256** of the hook **238** contacts a surface of the catch engagement **228**. The outer surface **256** of the hook **238** and the surface of the catch engagement **228** are facing surfaces that offset the angular force applied on the primary sear **206** by the actuator **204** from the firing spring **244**. More specifically, and as will be appreciated, the actuator **204** is under a linear force, e.g., along the axis **212** from the firing spring **244**. This force is transferred from the actuator **204** to the primary sear **206**, at the interface of the leading surface **214** of the actuator and the first protrusion **224** of the primary sear **206**. This force imparts a rotational force on the primary sear **206**, e.g., that would cause the primary sear **206** to rotate clockwise about the first pivot **230**. This rotational force is offset, at least partly, by the interface between the hook **238** and the catch engagement **228**.

(36) FIG. **2D** shows a firing configuration of the trigger assembly **200**. Specifically, in FIG. **2D**, a user has engaged the trigger **106**, causing the trigger **106** to pivot clockwise about the third pivot **236**, generally in the direction shown by the arrow **258**. With continued movement of the trigger **106** in the direction of the arrow **258**, the trigger extension **243** contacts an underside of a trailing arm **260** of the secondary sear **208**. The trigger extension **243** imparts a rotational force on the secondary sear **208**, causing the secondary sear **208** to rotate in a clockwise direction about the second pivot **234**. This rotation, in turn, causes the secondary sear surface **232** to rotate relative to the primary sear **206**, to the position of FIG. **2D**. In this position, the secondary sear **208** no longer supports the primary sear **206** in the cocked position. With the secondary sear **208** no longer in position to counteract the force applied on the primary sear **206** by the actuator **204**, the primary sear **206** rotates about the first pivot **230**, e.g., in a clockwise direction, such that the actuator **204** is allowed to advance, under the biasing force, in the firing direction. The biasing force, unchecked

by the primary sear **206**, causes the actuator **204** to advance with sufficient force to displace the valve stem **246**, thereby causing compressed air to rush through the barrel **102** to fire the air gun **100**.

(37) FIG. 2D also shows an optional second trigger extension **245**. Like the trigger extension **243**, the second trigger extension **245** may be a protrusion that extends upward (in the orientation of FIG. 2D). For example, the trigger extension **243** may be a columnar member having a distal end extending a height above a portion of the trigger **106**, e.g., above the trigger surface **242**. Like the trigger extension **243**, the second trigger extension **245** may be positioned to contact the secondary sear **208** when the trigger **106** is pulled, e.g., to cause the secondary sear **208** to disengage the primary sear **206**. In the example, the second trigger extension **245** is relatively closer to the pivot **236** about which the trigger **106** pivots. In examples, a height of the second trigger extension **245** may also be adjustable to alter aspects of the trigger assembly **200**. For example, the heights of the trigger extension **243** and the second trigger extension **245** may cooperate to adjust aspects of use of the trigger assembly **200**. For example, the pulling the trigger **106** as described may cause the second trigger extension **245** to first contact the secondary sear **208** and begin rotation of the secondary sear **208**, e.g., about the pivot **234**, and then the first trigger extension **243** may contact the secondary sear **208** to continue rotation of the secondary sear **208**. In examples, the use of the second trigger extension **245**, with the trigger extension **243**, may allow for a different “feel” to a user pulling the trigger **106**.

(38) As shown in FIG. 2D, the primary sear **206** rotates to a position in which the secondary sear surface **232** of the secondary sear **208** is in front of the leading end **220** of the primary sear **206**, that is, no longer engaging, as discussed above. However, the surface **256** of the hook **238** of the catch **210** and the engagement surface **228** of the primary sear may remain in contact, e.g., as the catch **210** rotates with the primary sear **206**.

(39) As noted above, the valve stem **246** is biased to an extended position shown in FIG. 2A. Under sufficient force, as just detailed, the valve stem **246** deflects (in the firing direction), causing air to rush into and through the barrel **102**, e.g., via the conduit **202** to fire the air gun **100**. Once fired, the biasing force on the valve stem **246** causes the valve stem **246** to return to the extended position, generally as shown in FIG. 2E. More specifically, FIG. 2E shows an automatic cocking configuration. In this configuration, the valve stem **246** may exert a sufficient force on the actuator **204** to cause the actuator to return toward the firing spring **244** and at least partially compress the firing spring **244**. In some examples, the actuator **204** may also or alternatively be forced toward the firing spring **244** by compressed air. For example, some of the compressed air that passes through the conduit **202** to fire the air gun **100** may instead pass through an opening in a direction toward the actuator **204**. Although not visible in FIG. 2E, a clearance may exist around the valve stem **246** through which some of the compressed air used to fire the air gun **100** may pass. In some examples, to facilitate sufficient force by this “back flow” air, the conduit **202** may be sized to provide air at a pressure and/or volume sufficient to both fire the air gun **100** and force the actuator **204** rearward, as shown in FIG. 2E. In examples, the force applied by the valve stem **246** and/or the “back flow” air on the actuator mimics the cocking force applied along the direction of the arrow **248** to cock the air gun **100**, discussed above in connection with FIG. 2B. Like in the cocking configuration of FIG. 2B, as the actuator **204** is forced in the direction of the arrow **248**, against the force of the firing spring **244**, the engagement surface **218** of the actuator **204** contacts the second protrusion **226** of the primary sear **206**, causing the primary sear **206** to rotate in a counterclockwise direction. Also as in the example of FIG. 2B, the first protrusion **224** at the leading end **220** of the primary sear **206** extends upward, into a travel path of the actuator **204**, as the primary sear rotates.

(40) Unlike the configuration of FIG. 2B, however, in the configuration of FIG. 2E the trigger **106** remains pulled, such that the trigger extension **243** continues to contact the secondary sear **208**, and retain the secondary sear **208** in the rotated position of FIG. 2C, e.g., such that the secondary sear

208 does not support the primary sear **206** from rotating in the clockwise direction. Instead, in the configuration of FIG. 2E, as shown in a highlighted portion **262**, the hook **238** of the catch **210** disengages from the primary sear **206** as the primary sear **206** rotates, such that a distal end of the hook **238**, including the surface **256**, is disposed in a notch **264** of the primary sear **206**. The notch **264** may be formed in the primary sear **206** proximate the catch engagement **228**. In more detail, and as shown in the highlighted portion **262**, a downward-facing (e.g., bottom) surface of the hook **238** contacts an upward-facing (e.g., top) surface of the catch engagement **228**, defined at least partially by the notch **264**.

(41) Once the actuator **204** contacts (and compresses) the firing spring **244**, the biasing force of the firing spring **244** will bias the actuator **204**, in the firing direction. As just discussed in connection with FIG. 2E, when the actuator **204** is forced back by the valve stem **246**, the primary sear **206** pivots such that the first protrusion **224** (at the leading edge **220** of the primary sear **206**) is in a path of travel of the actuator **204**. As shown in FIG. 2F, which is similar to the example of FIG. 2C discussed above, in this arrangement the leading surface **214** of the actuator **204**, under the force of the firing spring **244**, contacts the first protrusion **224** to maintain the actuator **204** against the spring force of the firing spring **244**. Unlike in the example of FIG. 2C, in FIG. 2F the secondary sear **208** is not positioned such that the secondary sear surface **232** is disposed under the leading end **220** of the primary sear **206**, e.g., because the trigger is still pulled. Accordingly, the secondary sear **208** does not resist motion of the primary sear **206**. Instead, the distal end of the hook **238**, including the surface **256**, is disposed in the notch **264** of the primary sear **206** to resist the force biasing the primary sear **206** to rotate, generally shown at the highlighted portion **266** in FIG. 2F.

(42) In the example of FIG. 2F, the actuator **204** is retained in the cocked position, but the trigger is already “pulled.” FIGS. 2G and 2H show the progression of resetting the trigger, e.g., to a normal or unpulled orientation, while maintaining the actuator **204** in the cocked and ready-for-firing position. Conceptually, FIGS. 2G and 2H illustrate steps in a “hand-off” of the force that counters the biasing force resulting from the firing spring **244** from the hook **238** (of the catch **210**) to the secondary sear **208**.

(43) In FIG. 2G, the trigger has been released and moves toward the normal, unpulled position. For example, a trigger return spring **268** may bias the trigger **106** to rotate counterclockwise in the example orientation of FIG. 2G. Thus, when pulling the trigger **106** to fire the trigger assembly **200**, the user will overcome the biasing force of the trigger return spring **268**. Also in FIG. 2G, the biasing of the trigger **106** in the counterclockwise direction causes the catch **210** to rotate in the counterclockwise direction. Specifically, the trigger surface **242** of the trigger **106** contacts the trigger engagement surface **240** of the catch, which in turn forces the counterclockwise rotation of the catch **210**. This interaction is generally illustrated at first highlighted portion **270** in FIG. 2G. As will be appreciated, continued counterclockwise rotation of the catch **210** will cause the hook **238** to disengage from the notch **264** of the primary sear **206**.

(44) As also illustrated in FIG. 2G, in addition to biasing the catch **210**, the counterclockwise rotation of the trigger **106**, as the trigger returns to the unpulled or normal position, will also cause the trigger extension **243** to rotate counterclockwise. When the trigger **106** was in the pulled position, the trigger extension **243** was maintaining the secondary sear in a clockwise position, e.g., against a biasing force of a secondary sear spring **272**. In the clockwise position, the secondary sear **208** is unable to support the primary sear **206**, as discussed further herein. Releasing the trigger **106** by allowing the trigger **106** to rotate counterclockwise, also allows the secondary sear **208** to rotate (also counterclockwise in the example shown). More specifically, the movement of the trigger extension **243** allows for a corresponding movement of the secondary sear **208**, under the force of the secondary spring sear **272**. This movement of the secondary sear **208** positions the secondary sear surface **232** under the primary sear **206**. The highlighted portion **274** of FIG. 2G shows the movement of the trigger extension **243** and the secondary sear **208**, with the secondary sear surface **232** in a position to support the primary sear **206**.

(45) FIG. 2H shows continued rotation of the trigger **106** to a normal or ready-for-firing position. More specifically, in FIG. 2H, the biasing force of the trigger return spring **268** has continued rotation of the trigger **106** in the counterclockwise direction to the normal position. In the normal position, a trigger stop **276** resists continued (counterclockwise) rotation of the trigger **106**. The trigger stop **276** may be adjustable, e.g., to facilitate setting of a preferred orientation of the trigger **106** in the normal position. In examples, the trigger stop **276** can include a set screw or the like, which is fixed relative to the air gun **100**.

(46) The continued rotation illustrated by FIG. 2H has also caused the trigger surface **242** to continue to interface with the trigger engagement surface **240**, to continue counterclockwise rotation of the catch **210**. Specifically, the catch **210** is rotated to a position at which the hook **238** disengages from the notch **264** of the primary sear **206**. Accordingly, in the example of FIG. 2H, the catch **210** no longer counters the biasing force on the primary sear **206** applied by the actuator **204**. Instead, the secondary sear **208**, which is now positioned under the primary sear **206**, counters this biasing force. As will be appreciated, the configuration of FIG. 2H corresponds to the cocked and ready for firing configuration of FIG. 2C. Thus, FIGS. 2D-2H illustrate firing and automatic re-cocking of the trigger assembly **200**, without any user interaction beyond pulling (and subsequently releasing) the trigger.

(47) In the example configurations of FIGS. 2G and 2H, the return of the trigger **120** to the unbiased position is caused by the trigger return spring **268**. The interface of the trigger surface **242** with the trigger engagement surface **240** causes movement of the catch **210** and thus disengagement of the catch **210** from the primary sear **206**. As noted above, the trigger engagement surface **240** is cammed to facilitate a desired release of the catch **210** from the primary sear **206**. In the example, the trigger engagement surface **240** has an arcuate surface, although, and as will be appreciated, modifications to the surface can result in varied disengagement times and/or movement patterns relative to return of the trigger **106** to the neutral or non-actuated position.

(48) The subject matter described above is provided by way of illustration only and should not be construed as limiting. Furthermore, the claimed subject matter is not limited to implementations that solve any or all disadvantages noted in any part of this disclosure. Various modifications and changes may be made to the subject matter described herein without following the examples and applications illustrated and described, and without departing from the spirit and scope of the present invention, which is set forth in the following claims.

Claims

1. A trigger assembly comprising: an actuator configured to move along a longitudinal axis, the actuator having a leading surface and a trailing surface; a primary sear configured to rotate about a first pivot, the primary sear including a first protrusion proximate a leading end of the primary sear and a second protrusion proximate a trailing end of the primary sear, the primary sear further comprising a catch engagement surface proximate the trailing end of the primary sear; a catch configured to rotate about a second pivot spaced from the first pivot; a trigger configured to pivot about the second pivot relative to the catch; and a secondary sear configured to rotate about a third pivot, wherein: with the trigger assembly in a cocked configuration, the first top protrusion of the primary sear contacts the leading surface of the actuator to retain the actuator in a cocked position and the secondary sear contacts the primary sear to resist pivoting of the primary sear, in response to rotation of the trigger about the second pivot to a firing position, the trigger contacts the secondary sear, causing the secondary sear to rotate about the third pivot to facilitate movement of the primary sear about the first pivot, the movement of the primary sear about the first pivot causing the first top protrusion of the primary sear to release the actuator, the release of the actuator causes the actuator to move along the barrel axis in a first direction to fire the air gun, after firing of the air gun, the actuator moves in a second direction along the barrel axis to contact the second

protrusion of the primary sear, causing the primary sear to pivot about the primary sear to a re-cocking position in which the catch engagement surface of the primary sear engages the catch and the catch retains the primary sear in a position to contact the leading surface of the actuator with the first protrusion of the primary sear, and releasing the trigger with the catch engaged causes, in order, the secondary sear to return to a position that engages the primary sear and causes the catch to disengage from the catch engaging surface, thereby placing the air gun in the cocked configuration.

2. The trigger assembly of claim 1, wherein: the trigger comprises a trigger blade; the catch comprises a cammed surface; and releasing the trigger causes the trigger blade to engage with the cammed surface to cause the catch to disengage from the catch engaging surface of the primary sear.

3. The trigger assembly of claim 1, wherein: the actuator comprises an engagement surface offset from the trailing end of the actuator; and the engagement surface contacts the second protrusion of the primary sear.

4. The trigger assembly of claim 1, further comprising: a biasing member biasing the trigger to a neutral position, wherein, in the neutral position, a trigger blade biases the catch out of engagement with the primary sear.

5. The trigger assembly of claim 4, further comprising: a trigger stop configured to retain the trigger in the neutral position against a force of the biasing member.

6. The trigger assembly of claim 4, further comprising a trigger extension extending from the trigger, the trigger extension being adjustable to configure a distance between the trigger extension and the secondary sear.

7. The trigger assembly of claim 1, further comprising: a biasing member biasing the secondary sear into a position to contact the primary sear to retain the hammer in the cocked position.

8. The trigger assembly of claim 1, wherein the actuator comprises a hammer.

9. The trigger assembly of claim 1, further comprising: a firing spring disposed along the barrel axis, proximate the trailing surface of the actuator, wherein the trailing surface of the actuator is held against a biasing force of the firing spring in the cocked position; and a valve stem disposed along the barrel axis, proximate the leading surface of the actuator, the valve stem configured to provide a return force on the actuator to cause the actuator to move in the second direction along the barrel axis.

10. An air gun comprising: a barrel; a compressed air source in fluid communication with the barrel; a valve configured to selectively allow compressed air from the compressed air source into the barrel; and a trigger assembly configured to selectively open the valve to pass compressed air from the compressed air source to the barrel to fire a projectile, the trigger assembly comprising: an actuator biased by a firing spring toward the barrel along a longitudinal axis; a primary sear configured to rotate about a first pivot; a secondary sear configured to rotate about a second pivot spaced from the first pivot; a catch configured to rotate about a third pivot spaced from the first pivot and the second pivot; and a trigger configured to pivot about the third pivot, relative to the catch, wherein: the secondary sear holds the primary sear in a first position contacting the actuator to retain the actuator in a cocked position against a biasing force of a firing spring with the trigger in a normal position, the catch holds the primary sear in the first position contacting the actuator to retain the actuator in a re-cocking position against a biasing force of the firing spring with the trigger in a pulled position, and returning the trigger from the pulled position to the normal position with the actuator in the re-cocking position causes, in order, the secondary sear to engage the primary sear and the catch to disengage from the primary sear.

11. The air gun of claim 10, further comprising: a valve stem associated with the valve and disposed proximate the barrel, the valve stem configured to provide a return force on the actuator to cause the actuator to move toward the firing spring, to the re-cocking position.

12. The air gun of claim 11, wherein: with the trigger assembly in the cocked configuration, a first

protrusion of the primary sear contacts a leading surface of the actuator to retain the actuator in the cocked position and the secondary sear contacts the primary sear to resist pivoting of the primary sear about the first pivot, in response to rotation of the trigger about the third pivot to the pulled position, the trigger contacts the secondary sear, causing the secondary sear to rotate about the second pivot to facilitate movement of the primary sear about the first pivot, the movement of the primary sear about the first pivot causing the first protrusion of the primary sear to release the actuator, the release of the actuator causes the actuator to move along the barrel axis in a first direction to contact the valve stem and fire the air gun, after firing of the air gun, the actuator moves in a second direction along the barrel axis to contact a second protrusion of the primary sear, causing the primary sear to pivot about the first pivot to a position in which a catch engagement surface of the primary sear engages the catch and the catch retains the primary sear in a position to contact the leading surface of the actuator with the first protrusion of the primary sear, retaining the actuator in the re-cocking position, and releasing the trigger with the catch engaged causes, in order, the secondary sear to return to a position that engages the primary sear and causes the catch to disengage from the catch engaging surface, thereby placing the air gun back in the cocked configuration.

13. The air gun of claim 11, wherein: the trigger comprises a trigger blade; the catch comprises a cammed surface; and releasing the trigger causes the trigger blade to engage with the cammed surface to cause the catch to disengage from the catch engaging surface of the primary sear.

14. The air gun of claim 13, further comprising: a biasing member biasing the trigger to a neutral position, wherein, in the neutral position, a trigger blade biases the catch out of engagement with the primary sear.

15. The air gun of claim 12, wherein: the actuator comprises an engagement surface offset from the trailing end of the actuator; and the engagement surface contacts the second protrusion of the primary sear.
